Final exam prep

One linked site. Sheets 1–3 are the midterm guides you already know; the new material (diesel engines, gas turbines, combined cycles, cooling towers) and the full set of past finals are added. Every solution has the given data, a sketch or diagram, the property lookups, the equations with numbers, and the final answer. Tick “I can solve this” as you go; progress is kept in this browser.

1 · Sheet 1 · Economics

Load curves, all the factors, cost per kWh.

0 / 5 done · Open →

2 · Sheet 2 · Steam cycles

Rankine, reheat, heaters, condition line, throttle governing, with lookup recipes.

0 / 11 done · Open →

3 · Sheet 3 · Steam generators

Lecture 5 plus drum, furnace, circulation, economizer, superheater.

0 / 9 done · Open →

4 · Sheet 4 · Diesel engines

Formula sheet, the repeated six-cylinder test (Q7), heat balances, supercharging.

0 / 15 done · Open →

5 · Sheet 5 · Gas turbines

Brayton cycle with losses, intercooling, reheat, regeneration; waste-heat boiler.

0 / 4 done · Open →

6 · Sheet 6 · Combined cycles & cogeneration

Gas + steam plants, dual-pressure steam generator, back-pressure turbine, waste-heat distillation.

0 / 4 done · Open →

7 · Sheet 7 · Cooling towers

Dry and wet towers, psychrometrics, Merkel number, Kelly's fill, L/G sweeps.

0 / 4 done · Open →

Finals · all four papers

Fall 2023–2026 solved in exam format, including the combined cycle and the dry cooling tower.

0 / 9 done · Open →

Practice · new problems

Exam-style problems with hints, answers and guided solutions.

0 / 6 done · Open →

What the past finals actually ask

- **Every paper:** an open-heater or multi-heater steam cycle with flow diagram, h–s or T–s diagram and efficiency (Sheet 2 Q6 and Q5).
- **2024, 2025, 2026:** the **same** six-cylinder diesel test (Sheet 4 Q7). If you learn one thing, learn this one.
- **Economics** (Sheet 1) or the **boiler drum** (Sheet 3 Q1) as Q1; the economizer and air preheater (Sheet 3 Q6) in 2023.
- **One "late-course" question:** combined cycle (2023) or a dry cooling tower with a surface condenser (2024).

Exam-day answer layout

1. Given / Find
2. Assumptions (steady flow, negligible KE/PE, isentropic pumps, TTD, stack temperature…)
3. Sketch: flow diagram with state numbers, then T–s or h–s (or p–V for engines)
4. Property table: state, p, T, h, s, x, with the table used
5. Steps: formula first, then numbers, then units
6. Boxed answer and a sanity check (efficiency below Carnot, quality between 0 and 1, heat balance closes)

Read this: where I derived or flagged things

- Sheet 4: your notebook solves Q1–Q6, Q8 and Q12. **Q7 (the repeated exam question)**, Q9–Q11 and Q13–Q15 had no solution, so I derived them. Two notebook numbers differ from mine and are flagged